

Ayush Ghatak

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Profile Summary

Data Science graduate specializing in machine learning, deep learning, optimization, and statistical modeling, building predictive models and scalable systems with applications in healthcare analytics and computer vision.

Experience

AI Research Intern, IAI, TCG Crest

June 2025 – April 2026

- Designed large-scale Kidney Exchange optimization using graph models, column generation, and branch-and-price on UNOS-like synthetic data; scaled to 10K patients with 2400s runtime ($\approx 31\%$ faster than prior work).
- Reproduced fairness-aware models and designed heuristics for scalability; observed 10–15% efficiency loss (Price of Fairness) and predicted the trade-off using XGBoost (MAE: 2.45%, R^2 : 0.90).
- Implemented dynamic matching framework FUTUREMATCH and ML-based predictive modeling and evaluation: XGBoost (group-based split, R^2 : 0.96, MAE: 3.3%) and Ridge Regression (temporal split, R^2 : 0.84, MAE: 5.2%).

Advanced Analytics Intern, PharmaQuant Insights Pvt. Ltd.

May 2025 – June 2025

- Analyzed 6–8 clinical trials using survival analysis and meta-analysis in R to support treatment evaluation.
- Generated pooled estimates (OS, PFS, ORR) to inform data-driven clinical and business decision-making.
- Applied the PICO framework to standardize research interpretation and improve consistency across studies.

Technical Skills

Programming Languages & Tools: Python, SQL, R, Power BI, Excel

Libraries: ML/DL: pandas, NumPy, sklearn, tensorflow, pytorch | **Graphs/Stats:** NetworkX, statsmodels, SciPy | **Vision/Data Visualization:** OpenCV, Matplotlib, Plotly | **NLP:** NLTK, SpaCy

Education

M.Sc. Data Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 9.01 2024-2026

B.Sc. Computer Science, St. Xavier's College (Autonomous), Kolkata — CGPA: 8.73 2021-2024

Class XII (CBSE), Notre Dame School, Delhi — 94.4% 2021

Major Projects

Drone-Based 3D Reconstruction Optimization (Computer Vision + Optimization) [github link](#)

Tech Stack: Python, cv2, NumPy, SciPy, PyProj, pandas, plotly, Optimization Algorithms

- Built data-driven 3D reconstruction pipeline on 176 high-resolution images using feature engineering (SIFT) and statistical filtering (Lowe's ratio test, RANSAC).
- Formulated reprojection error as objective function; reduced error from 18.71 to 2.91 pixels ($\approx 84\%$) using BFGS optimization.
- Improved spatial alignment by 94.6% via geospatial transformation and distance-based optimization (Newton's method).
- Achieved final reconstruction error of 0.23 m through cost minimization (Fibonacci search).

Hospital Readmission Prediction & Patient Segmentation (Healthcare ML) [github link](#)

Tech Stack: Python, sklearn, XGBoost, pandas, seaborn

- Built end-to-end ML pipeline for 30-day readmission prediction using clinical and demographic data.
- Performed data cleaning, feature engineering, and imputation.
- Trained Logistic Regression, Random Forest, and XGBoost; XGBoost performed best (ROC-AUC: 0.6565) and identified key risk drivers.
- Applied post-model K-means clustering for patient segmentation to identify high-risk groups and optimize care.

Achievements & Activities

- Secured first place in PowerQuest Hackathon 2025 (HEOR Analytics), PharmaQuant Insights Pvt. Ltd.
- Completed the "Power BI for Beginners" certification on Simplilearn, gaining hands-on experience in data modeling, dashboard design, and interactive data visualization.